Expected Free Energy Planning with Learned External-Camera Measurement Reliability*

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Abstract. A robot planning under uncertainty must trade goal progress against whether it will be able to localize itself effectively at potential future positions. We study this problem in Gazebo simulation for a mobile robot localized by a fixed external camera, where detector reliability varies with viewpoint, occlusion, and position in the image. The controller maintains a Gaussian pose belief. A global finite-horizon Expected Free Energy route solve optimizes a bounded unicycle-command sequence, and a local tracker follows the resulting waypoints. We fit a Gaussian process to the confidence scores of a trained object detector, giving a spatial estimate of detector reliability over the workspace. In a 40-run Gazebo campaign over four tasks and five seeds, the planner using this state-dependent covariance reaches all 20 goals without leaving the traversable region. The uniform-covariance planner reaches 15/20 goals and leaves the region in the other five runs, all on the camera-poor task. The state-dependent planner also has lower position-estimation error on the three tasks where the planners select different routes and comparable error on the control task. In this simulation, the learned covariance changes route selection on those three tasks, eliminates region exits on the camera-poor task, and reduces localization error when the selected routes differ.

Keywords: Active inference · robot navigation · external camera localization · Gaussian processes

1 Introduction

Mobile robots are increasingly used in industrial settings such as warehouses, terminals, and production facilities, where they automate internal transport between storage, picking, and delivery locations [7]. In these structured indoor environments, external sensing is an attractive alternative or complement to fully onboard perception. Fixed external cameras can support localization from

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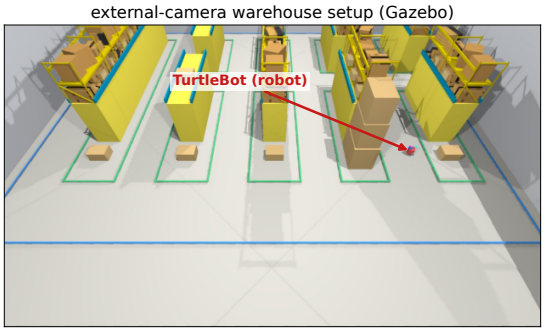


Fig. 1. Simulated fixed-camera warehouse setup used in the experiments.

different viewpoints, reduce the sensing burden on the robot itself, and provide global scene context [6,3].

We study finite-horizon belief-space route planning for a mobile robot observed by a fixed external camera with known pose (Fig. 1). The robot moves on a planar ground surface in a static environment, but its pose is not directly observed by the controller. Instead, the camera image is processed by a detector to obtain an image-space robot observation, and a calibrated homography relates this observation to the ground plane. The controller therefore acts on a belief, represented as a probability distribution over the robot pose, obtained by predicting the state with the motion model and correcting it with camera-derived measurements [16,15].

The global route solve ranks velocity-command sequences over a finite horizon by their expected free energy (EFE) [13,17]. The goal-directed risk term penalizes disagreement between predicted and preferred future observations, while the ambiguity term captures the conditional uncertainty of future camera-derived observations given predicted robot states [11]. If camera updates become weak or intermittent, the robot must rely more strongly on prediction and the state estimate can become less reliable. This can lead to collisions or loss of localization.

Practical external-camera localization systems exhibit state-dependent measurement reliability due to occlusion, image-boundary effects, lighting variation, detector failures, and calibration errors. These effects are difficult to capture completely with a manually designed geometric model alone. We model the detector confidence as a latent field over the workspace using a Gaussian Process (GP) that retains predictive uncertainty in regions with limited data [12,8].

We test whether accounting for predicted observation covariance can make the planner prefer routes that are expected to provide more reliable camera-derived state updates, rather than only geometrically short routes. The reliability-blended covariance inflates the predicted belief covariance in camera-poor regions, which raises the no-go clearance cost of an occluded route and can make an observable route preferable.

2 Problem Statement

We consider a mobile ground robot navigating in a known planar workspace $\mathcal{X} \subset \mathbb{R}^2$ while localized from a fixed external camera. A floor-plan map specifies the traversable region $\mathcal{X}_{\text{trav}} \subset \mathcal{X}$, which contains all positions available for robot motion. The robot starts from an initial pose belief and must reach the goal region $\mathcal{X}_{\text{goal}} = \{p \in \mathcal{X}_{\text{trav}} : \|p - p^\star\| < r_{\text{goal}}\}$, where $p^\star$ is the goal position.

2.1 System specification

The robot pose at time step k is $s_k = [x_k, y_k, \theta_k]^\top$, with planar position $p_k = [x_k, y_k]^\top$ and heading θ_k. The commanded input is $u_k = [v_k, \omega_k]^\top \in \mathcal{U}$, with bounded translational and angular velocity commands. The executed input may differ from the command. We write $\tilde{u}_k = u_k + \xi_k$, where ξ_k denotes actuation uncertainty from slip, delay, and low-level execution effects. The executed motion follows the unicycle model

$$s_{k+1} = f(s_k, \tilde{u}_k), \qquad f(s_k, \tilde{u}_k) = \begin{bmatrix} x_k + \tilde{v}_k \Delta t \cos\theta_k \\ y_k + \tilde{v}_k \Delta t \sin\theta_k \\ \theta_k + \tilde{\omega}_k \Delta t \end{bmatrix}.$$

The camera is modelled as a calibrated perspective camera with known intrinsics and extrinsics. Restricting the perspective projection to the ground plane gives the ground-plane homography [5]

$$\lambda \begin{bmatrix} \alpha \\ \beta \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \qquad H = K_{\text{cam}} \begin{bmatrix} r_1 \; r_2 \; t_c \end{bmatrix},$$

where K_{cam} is the camera intrinsic matrix, r_1, r_2 are the first two columns of the camera rotation, t_c is the camera-frame translation, and λ is the projective scale. We write the forward map from ground position to image point as

$$h(p) = \nu(H[p^\top, 1]^\top), \qquad \nu(\bar{z}) = \begin{bmatrix} \bar{z}_1/\bar{z}_3 \\ \bar{z}_2/\bar{z}_3 \end{bmatrix}.$$

The inverse map used by the localization pathway is

$$h_{\text{bev}}^{-1}(y^{\text{cam}}) = \nu(H^{-1}[\alpha, \beta, 1]^\top),$$

which maps an accepted image-space detector point to a planar position estimate used for visualization and detector-error evaluation. The homography is treated as fixed after calibration. Figure 2 illustrates this calibrated relation.

2.2 Generative model and belief update

The agent represents the initial pose uncertainty by $\mathbb{P}(s_0) = \mathcal{N}(s_0; m_0, S_0)$. It models transition uncertainty with

$$\mathbb{P}(s_k \mid s_{k-1}, u_{k-1}) = \mathcal{N}(s_k; f(s_{k-1}, u_{k-1}), Q),$$

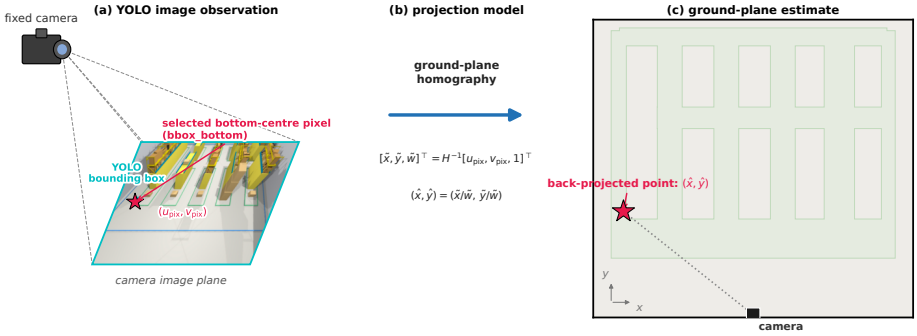


Fig. 2. The homography relates ground-plane position to image-space observations.

where Q summarizes the uncertainty that the agent assigns to effective disturbances and discretization. In the implementation this becomes the state-dependent discrete covariance $Q_d(\theta, v, \Delta t)$ derived in Appendix B.

The camera measurement model is written in image space:

$$\mathbb{P}(y_k^{\text{cam}} \mid s_k) = \mathcal{N}(y_k^{\text{cam}}; h(p_k), R_k).$$

The covariance R_k is the effective image-space uncertainty of the accepted detector point. The correction is evaluated as an image-space innovation through the homography linearization.

The controller maintains a Gaussian belief $q(s_k) \approx \mathcal{N}(s_k; m_k, S_k)$, where m_k is the pose estimate and S_k is the associated covariance. The belief is predicted with the motion model and corrected with the camera-derived measurement using an Extended Kalman Filter (EKF)-style linearization [15,16]. Let $\mathcal{D}_k$ denote the measurement and control history up to time k. Bayes' rule gives

$$\mathbb{P}(s_k \mid \mathcal{D}_k) = \frac{\mathbb{P}(y_k^{\text{cam}} \mid s_k)\,\mathbb{P}(s_k \mid u_{k-1}, \mathcal{D}_{k-1})}{\mathbb{P}(y_k^{\text{cam}} \mid u_{k-1}, \mathcal{D}_{k-1})} \approx \mathcal{N}(s_k; m_k, S_k).$$

With $p_k^- = [m_{k,x}^-, m_{k,y}^-]^\top$ and measurement Jacobian $G_k = \partial h/\partial s|_{s=m_k^-}$, the Kalman gain is

$$K_k = S_k^- G_k^\top \left(G_k S_k^- G_k^\top + R_k\right)^{-1}.$$

The Gaussian correction is

$$m_k = m_k^- + K_k \left(y_k^{\text{cam}} - h(p_k^-)\right), \qquad S_k = (I - K_k G_k) S_k^-.$$

If no detector point is accepted, the camera correction is skipped and the belief remains prediction-driven until the next accepted update. The prediction equations and linearization details are given in Appendix A.

Figure 3 shows the route-choice setting. The shorter route passes near shelf geometry where detector updates can become weak or intermittent, while the longer route remains in regions with more reliable external-camera measurements.

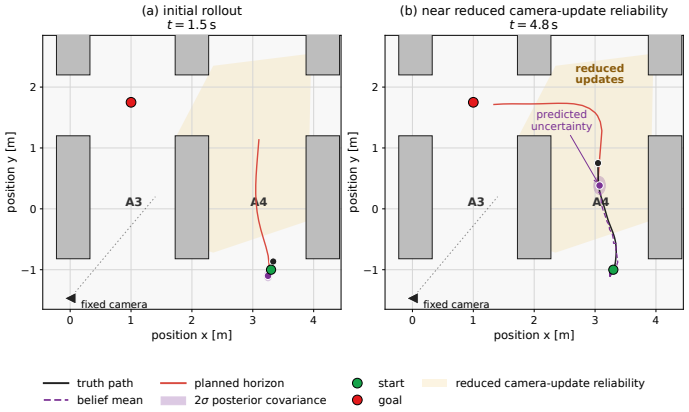


Fig. 3. Route-choice setting under a constant observation covariance (top-down view). The two snapshots show a constant-R_0 rollout leaving the start (a) and later approaching a region of reduced camera-update reliability (b), where the belief covariance grows.

3 Learned Camera-Update Reliability Model

The planner needs camera-measurement covariances at future states along candidate rollouts. These covariances cannot be observed at planning time, and a purely geometric camera model does not capture detector failures caused by occlusion, viewpoint, image location, or the trained detector itself. We therefore learn a detector-score-derived reliability proxy from detector outputs collected with the same fixed camera–detector setup used at runtime.

3.1 Detector-score target

During data collection, a YOLOv11n-seg detector processes the camera image and returns an image-space robot point together with a detector confidence score c_i [9]. The image-space point is the bottom centre of the selected robot bounding box. Appendix section C reports the detector training, validation metrics, and the offline localization accuracy of this bottom-centre point.

Let $p = [x, y]^\top$ denote planar position. Each sampled pose i contributes a detector-outcome score $c_i \in [0, 1]$. When a robot detection is selected, c_i is the confidence score of that selection. When no detection is selected, the sample is retained with $c_i = 0$. Scores from sampled robot headings at the same planar position are averaged before fitting.

3.2 Spatial GP reliability field

A Gaussian process interpolates from the sampled detector-score targets to query future positions while retaining uncertainty about the interpolation. The posterior mean gives the inferred detector-score field, and the posterior variance identifies regions where the field is weakly supported by data.

Before fitting, the averaged scores are clipped away from exactly 0 and 1 and transformed to logit space. We model the transformed target as $r_j = f_\rho(p_j) + \eta_j$,

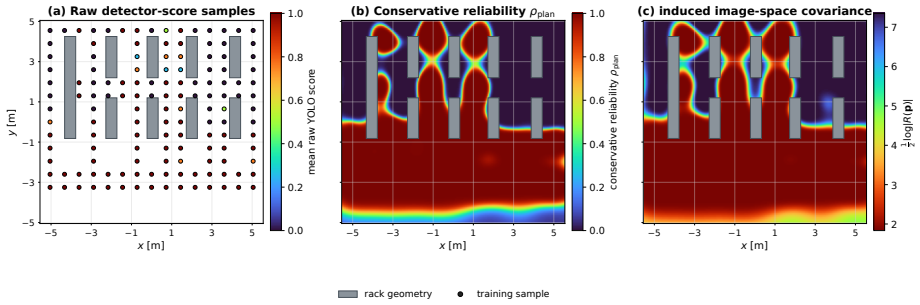


Fig. 4. Panel (a) shows detector confidence-score samples collected over the workspace. Panel (b) shows the conservative reliability field ρ_{plan} obtained from the fitted GP. Panel (c) shows the induced state-dependent image-space observation covariance.

with $\eta_j \sim \mathcal{N}(0, \sigma_n^2)$ and $f_\rho(p) \sim \mathcal{GP}(0, k_\rho(p, p'))$. The covariance function k_ρ is an RBF kernel with signal variance σ_f^2 and length scale ℓ_ρ, encoding the assumption that nearby floor positions have correlated detector-score reliability [14]. Let $\mu_\rho(p)$ and $\sigma_\rho(p)$ denote the posterior mean and standard deviation of the latent field at query position p.

The reliability map used for planning subtracts the posterior uncertainty from the mean before mapping back to $[0, 1]$, which lowers predicted reliability in weakly supported regions, as $\rho_{\text{plan}}(p) = \text{sigmoid}(\mu_\rho(p) - \beta_\rho \sigma_\rho(p))$, where $\beta_\rho \geq 0$ controls how strongly posterior uncertainty lowers the reliability used for planning.

3.3 Belief-aware covariance mapping

During planning, future robot position is represented by a predicted belief rather than a point estimate, and the predicted reliability is mapped to the effective observation covariance R^{eff} supplied to the planner by Eq. (2). For a predicted belief

$$q(s_{t+k} \mid \mathcal{D}_t) \approx \mathcal{N}(s_{t+k}; m_{t+k|t}, S_{t+k|t}),$$

let $m_{xy,t+k|t}$ and $S_{xy,t+k|t}$ denote its planar position marginal. The reliability map is queried at sigma points $p^{(i)}_{t+k|t}$ around this marginal, with weights w_i, and averaged:

$$\bar{\rho}_{t+k|t} = \sum_i w_i \rho_{\text{plan}}(p^{(i)}_{t+k|t}), \qquad \sum_i w_i = 1. \tag{1}$$

The scalar reliability is converted into an effective image-space observation covariance by interpolating between two measurement-information regimes. Let R_{vis} denote the covariance assigned to a reliable detector update and R_{miss} the covariance assigned to a weak or missed update, with $R_{\text{miss}} \succ R_{\text{vis}}$. In the experiments these are isotropic image-space covariances, $R_{\text{vis}} = r^2_{\text{vis}} I_2$ and $R_{\text{miss}} = r^2_{\text{miss}} I_2$. We blend in precision space,

$$\Lambda^{\text{eff}}_{t+k|t} = \bar{\rho}_{t+k|t} R^{-1}_{\text{vis}} + \left(1 - \bar{\rho}_{t+k|t}\right) R^{-1}_{\text{miss}}, \qquad R^{\text{eff}}_{t+k|t} = \left(\Lambda^{\text{eff}}_{t+k|t}\right)^{-1}. \tag{2}$$

The precision R^{-1} determines how strongly a measurement can correct the predicted belief. In the EKF correction, the innovation remains $y_k^{\text{cam}} - h(m_{xy,k}^-)$, as in Eq. (6). The score-conditioned covariance R_k is attached to this observed measurement update. At runtime, a missed detection skips the correction. R_{miss} is the planning model for the low information expected from weak or missed future observations.

4 EFE Route Planning

Route choice is determined by the global EFE-plus-no-go objective, solved once per episode to the goal from the current belief. The route is not re-solved online, and periodic replanning is left to future work.

4.1 Global route solve

At the start of a global planning phase, the planner optimizes a finite sequence of N_H bounded unicycle commands $U = (u_0, \ldots, u_{N_H-1})$, with $u_i \in \mathcal{U}$. The optimizer is initialized from initial routes derived from a lane network over the known traversable map. For each candidate sequence, the planner propagates the belief and the predicted camera observation distribution over the horizon. The optimized state sequence defines the route shape. CasADi assembles the differentiable finite-horizon objective and derivative information [1]. L-BFGS-B solves the resulting box-constrained nonlinear program over bounded controls [4], using the same initial routes in both conditions. The selected optimized state sequence is converted into fixed-spacing waypoints and executed by a bounded proportional unicycle tracker shared by both conditions. The tracker details are given in Appendix D.3.

4.2 Predictive belief and observation rollout

The rollout starts from the current Gaussian belief $q(s_0) \approx \mathcal{N}(s_0; m_0, S_0)$. Using the transition model from Section 2.2, a candidate command sequence U induces predicted beliefs

$$q(s_i \mid U) \approx \mathcal{N}(s_i; m_i, S_i), \qquad i = 1, \ldots, N_H.$$

The predicted image-space observation uses the projection function h from Section 2.1. After first-order linearization around the predicted mean,

$$q(o_i \mid U) \approx \mathcal{N}(o_i; \mu_i^o, \Sigma_i^o), \qquad \mu_i^o = h(m_{xy,i}),$$

with

$$\Sigma_i^o = G_i S_i G_i^\top + R_i^{\text{plan}}, \qquad G_i = \left.\frac{\partial h}{\partial s}\right|_{s=m_i}.$$

Here R_i^{plan} is the covariance assigned to the predicted camera observation at rollout step i. For C1, $R_i^{\text{plan}} = R_0$. For C2, it is the reliability-blended covariance defined in Equation 2.

4.3 Expected free energy objective

Candidate routes are ranked by the discounted finite-horizon objective

$$J(U) = \sum_{i=1}^{N_H} \gamma^{i-1} \left[w_{\text{risk}} J^{(i)}_{\text{risk}} + w_{\text{amb}} J^{(i)}_{\text{amb}} + J^{(i)}_{\text{nogo}} \right].$$

The risk term is the KL divergence from the predicted observation distribution to a preferred observation distribution associated with the goal. Let

$$\mathbb{P}^\star_i(o_i) = \mathcal{N}(o_i; \mu^\star_i, \Sigma^\star_i), \qquad \Delta_i = \mu^\star_i - \mu^o_i.$$

The preferred mean $\mu^\star_i$ is the image-space projection of the goal position. Its covariance is broad early in the horizon and tighter near the end, so early rollout states are not forced to match the final goal observation. The Gaussian risk term is

$$J^{(i)}_{\text{risk}} = \frac{1}{2} \left[\text{tr}\big((\Sigma^\star_i)^{-1}\Sigma^o_i\big) + \Delta_i^\top (\Sigma^\star_i)^{-1}\Delta_i - d_o + \log \frac{|\Sigma^\star_i|}{|\Sigma^o_i|} \right],$$

where d_o is the image-observation dimension.

The ambiguity term is the conditional entropy of the predicted observation under the joint Gaussian state-observation approximation. Let $\Gamma_i = S_i G_i^\top$ be the state-observation cross-covariance. The conditional observation covariance is $\Sigma^{o|s}_i = \Sigma^o_i - \Gamma_i^\top S_i^{-1}\Gamma_i$. The ambiguity term is

$$J^{(i)}_{\text{amb}} = \frac{1}{2} \left[d_o \log(2\pi e) + \log |\Sigma^{o|s}_i| \right].$$

This term uses the same predicted observation model as the risk term, including the condition-dependent camera-measurement covariance.

The planner uses the expected posterior state covariance associated with the predicted image-space observation,

$$S^+_i = S_i - \Gamma_i (\Sigma^o_i)^{-1} \Gamma_i^\top.$$

4.4 Uncertainty-inflated no-go term

The planner includes an uncertainty-inflated no-go term for known blocked workspace geometry. Writing $d_{\text{free}}(m_{xy,i})$ for the signed distance from the predicted position to the nearest no-go boundary, positive in free space, the clearance is

$$d_{\text{clr},i} = d_{\text{free}}(m_{xy,i}) - d_{\text{safe}} - \kappa \sqrt{\lambda_{\max}(S^+_{xy,i})},$$

where the last term inflates the boundary by κ standard deviations along the major axis of the planar position covariance. Lower predicted camera reliability therefore increases $S^+_{xy,i}$, reduces $d_{\text{clr},i}$, and makes routes near blocked geometry more costly. The penalty applied to $d_{\text{clr},i}$ is given in Appendix D.4.

5 Experiments

5.1 Setup and conditions

We evaluate on route-choice tasks in a Gazebo warehouse world adapted from the AWS RoboMaker Small Warehouse World package [2], hereafter the warehouse world. We use four start–goal pairs T_1–T_4, each with two routes, a short one through the rack aisles and a longer detour along an outer aisle. T_1 and T_2 start at the loading apron and end mid-aisle, where both routes stay well observed. T_3 runs from the west service lane to the upper end of the first rack aisle and approaches its goal through a camera-poor region. The control task T_4 shares T_3's start but ends lower in the same aisle, where camera detections are reliable throughout. The constant-covariance baseline (C1) uses a uniform camera covariance $R^{\text{eff}} = R_0 = R_{\text{vis}}$, i.e. it assumes a reliable detector update everywhere. The condition using reliability-blended covariance (C2) uses the GP reliability field from section 3 to construct the precision-blended covariance in Equation 2. The hierarchical controller uses a single global EFE solve over $N_H = 75$ steps at $\Delta t = 0.4\,\text{s}$, a 30 s lookahead that covers the full route, followed by the local waypoint tracker. The global solve is initialized with the same routes, generated from the known traversable map, in both conditions. The full runtime configuration and task coordinates are listed in subsection D.1.[1]

5.2 Single-task route choice

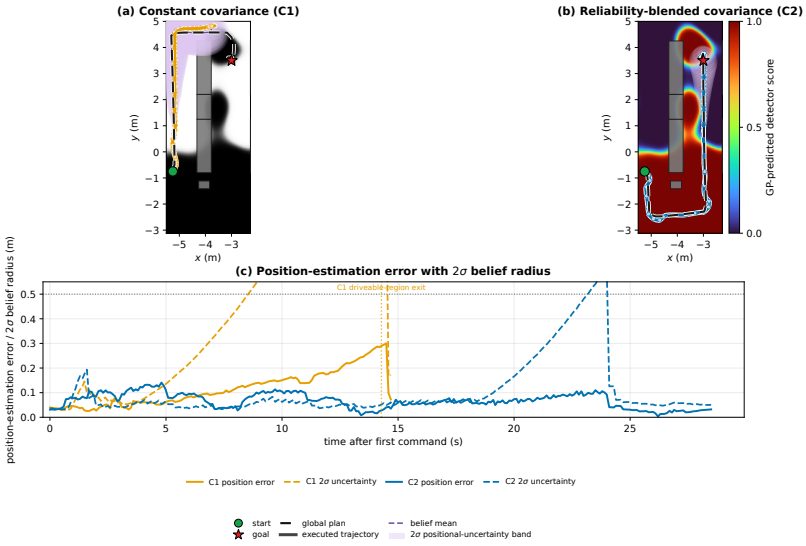


Fig. 5. Matched T_3 executions for seed 0. Panels (a,b) overlay the single global planned route (dashed) on the executed trajectory (solid). Panel (c) shows position-estimation error with the 2σ belief radius.

[1] Code and data: https://github.com/JoostLeliveld/iwai2026-camera-reliability-efe.

Figure 5 compares matched T_3 executions for the same seed with only the planner covariance changed. C1 takes the short route through the camera-poor region, where the belief receives no corrections, so its position uncertainty grows under the motion model alone and the estimate drifts from the truth as odometry error accumulates. C2 detours through the corridor where detections stay reliable and its error only starts growing once it has to cross the region the camera cannot see to reach the goal.

5.3 Multi-task comparison

The Gazebo execution campaign contains 40 runs [10]: four tasks, two planner conditions, and five seeds per condition. The seeds vary the simulator's command and encoder noise realizations and the initial belief. Clean success requires the robot to remain within 0.25 m of the goal for 2.0 s, and runs time out after 300 s from the first command.

Across the campaign, C2 reaches the goal cleanly in 20/20 runs and never leaves the traversable region, while C1 reaches the goal in 15/20 runs and leaves the traversable region in 5/20. All five C1 failures occur on the camera-poor T_3 route. Figure 6 shows the trajectories for all 40 runs.

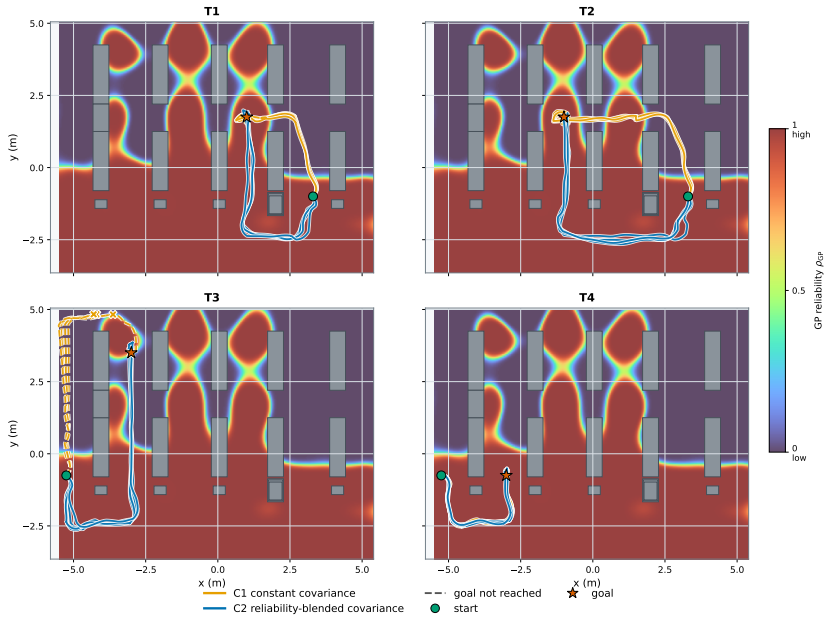


Fig. 6. Ground-truth trajectories across four tasks and five seeds per condition. Dashed trajectories with crosses did not reach the goal.

Position error is similar on T_1, T_2, and T_4, and separates on the camera-poor T_3 route (Table 1), where C1 accumulates roughly 50% more error with a spread three times wider.

Table 1. Run-level mean planar position-estimation error, as mean ± standard deviation over the five seeds (metres).

	T_1	T_2	T_3	T_4
C1	0.077 ± 0.003	0.080 ± 0.010	0.106 ± 0.017	0.069 ± 0.004
C2	0.072 ± 0.002	0.074 ± 0.002	0.072 ± 0.005	0.072 ± 0.002

6 Discussion

In T_3, reliability-blended covariance assigns larger predicted camera-update uncertainty to the occluded gap. This changes the predicted observation distribution used by the EFE risk and ambiguity terms and increases the clearance cost near blocked geometry. In the matched T_3 executions, C2 therefore selects the longer route through the corridor with more reliable camera detections, whereas C1 assumes the same camera-update covariance along both alternatives and commits to the shorter camera-poor route.

This route difference emerges with the EFE risk and ambiguity weights fixed at one, and the larger no-go weight held identical, in both conditions. The route preference is therefore not introduced by an explicit visibility reward or by condition-specific weighting, but by the reliability-blended camera-update covariance entering the predicted observation model and belief rollout. The no-go term converts the larger predicted covariance into a higher clearance cost within the same route objective. That term also makes the global route solve nonconvex and numerically demanding, so the current implementation depends on using the same set of initial routes in both conditions and remains computationally expensive.

6.1 Limitations

The learned reliability target is based on the raw YOLO detector score. This is not a direct measurement of metric localization error. Even with a high score, the selected bottom-centre pixel can be biased by the detector box, partial occlusion, image location and camera viewing geometry. These effects limit how accurately the EKF belief covariance can represent the true localization error.

The GP reliability field is an empirical component of the observation model. It introduces design choices that are not fixed by the robot dynamics or camera geometry alone, including the training target, kernel, uncertainty discount, precision blend, and covariance limits. The route choice is specific to the implemented camera, detector, world geometry, and EFE weights. The campaign in subsection 5.3 tests variation over tasks and random seeds, but it does not establish parameter-insensitive behaviour.

The current hierarchy also limits online adaptation. The global EFE solve is computationally expensive and is used to choose the route shape before local execution. The local waypoint tracker gives fast feedback motion, but it does not provide a new global route when the belief, detector behaviour, or workspace changes substantially during execution.

6.2 Future work

Future work should reduce sensitivity to modelling and numerical choices in the belief-space planner, in particular the interaction between the transition covariance Q, the camera-update covariance R, and the no-go calculation. Periodic global replanning from the current belief, with warm starts or a cheaper approximate solver, would reduce dependence on a single long-horizon rollout and let updated camera evidence, missed detections, and changing belief covariance affect the route during execution. Online updating of the reliability field from streaming detector outcomes would likewise reduce dependence on an offline map when lighting changes, objects move, or the detector drifts.

The same covariance interface could be tested with other sensing modalities, provided the sensor's measurement quality varies over space and can be mapped to a covariance. Examples include onboard visual feature quality, LiDAR return reliability, or radio-signal strength. Sensors with sharper loss-of-signal regimes may produce clearer failure cases and stronger route separation.

7 Conclusion

A Gaussian process fit to detector confidence scores was used to construct a spatially varying observation covariance for a finite-horizon Expected Free Energy route planner guided by a fixed external camera. Across four tasks with five seeds each, the condition using reliability-blended covariance more often selected the longer route with reliable camera detections, where the robot stayed detected and better localized. The matched constant-covariance baseline took the shorter occluded route more often, lost camera updates, and its degraded localization was associated with leaving the traversable region. The route difference emerged from the full route objective under identical initial routes, EFE weights, and no-go weights, with the camera-update covariance used by the planner as the only difference between conditions. The evidence remains specific to the fixed camera, detector, workspace, and planner configuration considered here. Physical-hardware validation remains future work.

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A Derivations for state estimation

State estimation is cast as recursive Bayesian filtering, alternating a motion-model prediction with a camera-derived correction [15,16]. Let

$$\mathcal{D}_{k-1} = \{y^{\text{cam}}_{1:k-1}, u_{0:k-2}\}$$

denote the measurement and control history before observing y^{cam}_k, and let

$$\mathbb{P}(s_{k-1} \mid \mathcal{D}_{k-1}) \approx \mathcal{N}(s_{k-1}; m_{k-1}, S_{k-1}).$$

The prediction step is

$$\begin{aligned}\mathbb{P}(s_k \mid u_{k-1}, \mathcal{D}_{k-1}) &= \int \mathbb{P}(s_k \mid s_{k-1}, u_{k-1})\, \mathbb{P}(s_{k-1} \mid \mathcal{D}_{k-1})\, \mathrm{d}s_{k-1} \\ &\approx \mathcal{N}(s_k; m_k^-, S_k^-). \end{aligned} \tag{3}$$

Using a first-order linearization of the unicycle model around m_{k-1},

$$m_k^- = f(m_{k-1}, u_{k-1}), \qquad S_k^- = F_{k-1} S_{k-1} F_{k-1}^\top + Q_{k-1},$$

with

$$F_{k-1} = \left.\frac{\partial f}{\partial s}\right|_{s=m_{k-1}, u=u_{k-1}}.$$

Here, Q_{k-1} is the transition covariance assumed by the filter and planner. It is a modelled uncertainty term, summarizing imperfect execution and residual process effects, and is kept fixed across the compared methods.

The camera correction uses the image-space likelihood

$$y_k^{\text{cam}} = h(p_k) + z_k, \qquad z_k \sim \mathcal{N}(0, R_k), \tag{4}$$

where $p_k = [x_k, y_k]^\top$ and h is the ground-plane-to-image projection defined in Section 2. Linearizing h around the predicted mean gives

$$h(p_k) \approx h(p_k^-) + G_k(s_k - m_k^-), \qquad G_k = \left.\frac{\partial h}{\partial s}\right|_{s=m_k^-},$$

where $p_k^- = [m_{k,x}^-, m_{k,y}^-]^\top$. Since the camera measurement depends only on planar position, the heading column of G_k is zero.

Bayes' rule gives

$$\begin{aligned}\mathbb{P}(s_k \mid \mathcal{D}_k) &= \frac{\mathbb{P}(y_k^{\text{cam}} \mid s_k)\,\mathbb{P}(s_k \mid u_{k-1}, \mathcal{D}_{k-1})}{\mathbb{P}(y_k^{\text{cam}} \mid u_{k-1}, \mathcal{D}_{k-1})} \\ &\approx \mathcal{N}(s_k; m_k, S_k).\end{aligned} \tag{5}$$

With Kalman gain

$$K_k = S_k^- G_k^\top \left(G_k S_k^- G_k^\top + R_k\right)^{-1},$$

the Gaussian correction is

$$m_k = m_k^- + K_k\left(y_k^{\text{cam}} - h(p_k^-)\right), \qquad S_k = (I - K_k G_k) S_k^-. \tag{6}$$

B Discretization of the nonlinear dynamics

The continuous-time kinematics of the unicycle, with additive zero-mean Gaussian disturbances acting on the actuation signals, read

$$\dot{x} = (v + w_v)\cos\theta\,, \tag{7}$$

$$\dot{y} = (v + w_v)\sin\theta\,, \tag{8}$$

$$\dot{\theta} = \omega + w_\omega\,, \tag{9}$$

where $s = [x, y, \theta]^\top$ is the pose, $u = [v, \omega]^\top$ collects the commanded forward and angular velocities, and w_v, w_ω are independent white-noise processes with power

spectral densities (PSDs) σ_v^2 and σ_ω^2. Stacking the disturbances in $w = [w_v, w_\omega]^\top$, the dynamics take the form $\dot{s} = f(s,u) + L(\theta)\, w$ with input map

$$L(\theta) = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix}, \tag{10}$$

so that the actuation noise is rotated into the world frame before it enters the state.

Discretizing over a step Δt around a frozen operating point (v_k, θ_k), the mean propagates through the Euler update $x_{k+1} = x_k + v_k \cos\theta_k\, \Delta t$, $y_{k+1} = y_k + v_k \sin\theta_k\, \Delta t$, and $\theta_{k+1} = \theta_k + \omega_k\, \Delta t$. The corresponding state-space PSD is $Q_c = L(\theta_k)\, \mathrm{diag}(\sigma_v^2, \sigma_\omega^2)\, L(\theta_k)^\top$. Integrating this PSD across the step while propagating it through the linearized flow $e^{F\tau}$, with $F = \partial f / \partial s$ evaluated at the operating point, gives the discrete process noise covariance

$$Q_d = \int_0^{\Delta t} e^{F\tau}\, Q_c\, e^{F^\top \tau}\, \mathrm{d}\tau\,. \tag{11}$$

For the unicycle the Jacobian

$$F = \begin{bmatrix} 0 & 0 & -v\sin\theta \\ 0 & 0 & v\cos\theta \\ 0 & 0 & 0 \end{bmatrix} \tag{12}$$

is nilpotent ($F^2 = 0$), so the matrix exponential truncates *exactly* to $e^{F\tau} = I + F\tau$. Substituting into Equation 11 and integrating term by term yields the closed form

$$Q_d = Q_c\, \Delta t + \tfrac{1}{2}\big(FQ_c + Q_c F^\top\big)\, \Delta t^2 + \tfrac{1}{3}\, FQ_c F^\top\, \Delta t^3\,. \tag{13}$$

Since F acts only on the heading column of Q_c, the higher-order corrections involve σ_ω^2 alone and encode how heading-rate uncertainty leaks into position over the step. Evaluating the products at (v_k, θ_k) and writing $c_\theta = \cos\theta_k$, $s_\theta = \sin\theta_k$ produces the final process noise covariance

$$Q_d = \begin{bmatrix} \sigma_v^2 c_\theta^2\, \Delta t + \frac{1}{3} v^2 s_\theta^2 \sigma_\omega^2\, \Delta t^3 & \sigma_v^2 c_\theta s_\theta\, \Delta t - \frac{1}{3} v^2 c_\theta s_\theta\, \sigma_\omega^2\, \Delta t^3 & -\frac{1}{2} v s_\theta\, \sigma_\omega^2\, \Delta t^2 \\ \sigma_v^2 c_\theta s_\theta\, \Delta t - \frac{1}{3} v^2 c_\theta s_\theta\, \sigma_\omega^2\, \Delta t^3 & \sigma_v^2 s_\theta^2\, \Delta t + \frac{1}{3} v^2 c_\theta^2 \sigma_\omega^2\, \Delta t^3 & \frac{1}{2} v c_\theta\, \sigma_\omega^2\, \Delta t^2 \\ -\frac{1}{2} v s_\theta\, \sigma_\omega^2\, \Delta t^2 & \frac{1}{2} v c_\theta\, \sigma_\omega^2\, \Delta t^2 & \sigma_\omega^2\, \Delta t \end{bmatrix}, \tag{14}$$

with the dependence on v_k left implicit. At $v = 0$ all corrections vanish, leaving the rank-one translational block $\sigma_v^2\, [c_\theta, s_\theta]^\top [c_\theta, s_\theta]\, \Delta t$ alongside $\sigma_\omega^2\, \Delta t$ on the heading. Keeping only the leading term recovers the first-order approximation $Q_d \approx L(\theta_k)\, \mathrm{diag}(\sigma_v^2, \sigma_\omega^2)\, L(\theta_k)^\top \Delta t$. The frozen operating point is consistent with the filter linearization and is accurate provided the heading turns little within a step. For large angular rates a smaller Δt is preferable to retaining further terms.

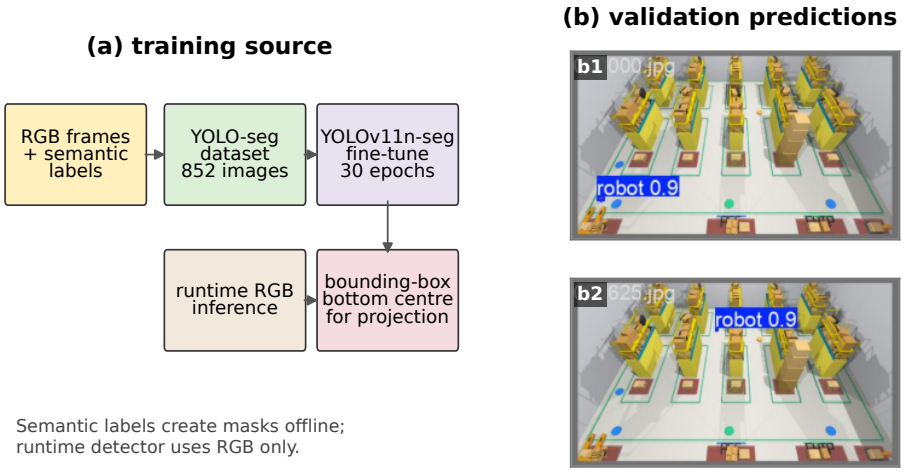


Fig. 7. YOLO training provenance and validation examples. The simulator semantic labels create the offline YOLO-seg masks, while the runtime detector uses only RGB images. The validation panel shows predictions from the fine-tuned detector on held-out simulated warehouse views.

C YOLO detector training and validation

The detector used for the warehouse experiments is a YOLOv11n-seg model [9] fine-tuned for the single class `robot`. The training set was generated in `warehouse_aws.world.sdf` from paired RGB images and simulator-native semantic segmentation labels. The accepted dataset contains 852 images, split deterministically into 683 training images and 169 validation images.

The model was fine-tuned for 30 epochs with image size 640, batch size 8, deterministic seed 0, and the training implementation's automatic optimizer setting. The runtime detector uses confidence threshold 0.05, IoU threshold 0.45.

The final validation row in `results.csv` reports box precision 0.982, box recall 0.889, box mAP50 0.938, and box mAP50–95 0.620. The corresponding mask metrics are precision 0.795, recall 0.769, mAP50 0.745, and mAP50–95 0.250.

The offline localization evaluation contains 912 frames and 647 thresholded YOLO detections. On the 645 detections that were also oracle-visible, the selected bottom-centre pixel residual is 5.95 px on average. Back-projecting the selected pixel with the same camera model gives a raw ground-plane error of 0.062 m mean (0.065 m median, 0.070 m RMSE, 0.096 m p90). The bottom-centre bounding-box point is only a proxy for the ground-contact point, so under the oblique camera view it carries a small systematic depth bias. A single fixed +0.05 m offset along the bird's-eye y-axis, chosen to centre the back-projected validation detections on the true ground positions, removes most of this bias: the calibrated ground-plane error is 0.049 m mean (0.046 m median, 0.058 m RMSE, 0.079 m p90). This is the same offset applied at runtime.

D Experiment details

D.1 Runtime configuration

Table 2 lists the locked runtime configuration shared by both conditions. C1 and C2 are identical except for the camera observation covariance R^{eff}, which is constant for C1 and GP-modulated for C2.

Table 2. Locked runtime configuration for the warehouse experiments.

Group	Setting
Tasks (start → goal, m)	$T_1 : (3.30, -1.00) \to (1.00, 1.75)$; $T_2 : (3.30, -1.00) \to (-1.00, 1.75)$; $T_3 : (-5.25, -0.75) \to (-3.00, 3.50)$; $T_4 : (-5.25, -0.75) \to (-3.00, -0.75)$
World / robot	`warehouse_aws.world.sdf`; $v_{\max} = 0.6\,\text{m/s}$; filter step $\Delta t = 0.25\,\text{s}$; robot collision radius 0.125 m
Global solve	solved once per run; horizon $N_H = 75$ at $\Delta t_{\text{g}} = 0.4\,\text{s}$ (30 s lookahead); L-BFGS-B, maxiter 60; two lane-network initial routes
Objective weights	$w_{\text{risk}} = 1.0$, $w_{\text{amb}} = 1.0$, $\gamma = 0.995$
Process noise	$\sigma_v = 0.012\,\text{m}\,\text{s}^{-1}/\sqrt{\text{s}}$; $\sigma_\omega = 0.05\,\text{rad}\,\text{s}^{-1}/\sqrt{\text{s}}$
Goal prior	image-space std annealed 50 → 12 px, power 0.9
No-go warning band	$w_{\text{nogo}} = 2000$, $w_{\text{near}} = 50$, safe distance 0.25 m, band $b = 0.05\,\text{m}$, $\kappa = 1.0$, $\epsilon = 0.01$
Detector	confidence 0.05, IoU 0.45, bounding-box bottom centre, masks off
Update gate	χ^2 innovation gate, 2 DOF, threshold 9.21 (99%)
Pixel noise scales	$r_{\text{vis}} = 2.5\,\text{px}$, $r_{\text{miss}} = 40\,\text{px}$
Command noise	slip mean/std 0.02/0.03, additive 0.005/0.02, correlation 0.85
Encoder noise	slip mean/std 0.01/0.02, additive 0.003/0.03, correlation 0.80

D.2 GP reliability artifact and covariance mapping details

The fitted GP artifact contains 139 heading-aggregated training positions on a 220×200 query grid over $[-5.5, 5.5] \times [-5.0, 5.0]\,\text{m}$. The aggregated detector-score target has mean 0.5971; before fitting it is clipped to $[10^{-4}, 1 - 10^{-4}]$. The RBF length scale is $\ell_\rho = 0.90\,\text{m}$, the GP noise variance is $\sigma_n^2 = 0.05$, the uncertainty discount is $\beta_\rho = 0.5$, and the minimum probability floor is 10^{-4}. The resulting conservative reliability map used for planning has minimum 10^{-4}, maximum 0.9999, and mean 0.5727.

D.3 Local waypoint tracker

The global solution found by EFE minimization is converted into a waypoint sequence $W = (w_1, \ldots, w_M)$, with $w_j \in \mathbb{R}^2$, using fixed spatial spacing along the predicted planar trajectory. During execution, the local layer tracks the active waypoint from the current belief mean $m_k = [m_{xy,k}^\top, m_{\theta,k}]^\top$. The active

waypoint defines

$$e_k = w_{\ell_k} - m_{xy,k}, \qquad \theta_k^{\mathrm{ref}} = \mathrm{atan2}(e_{y,k}, e_{x,k}), \qquad \psi_k = \mathrm{wrap}(\theta_k^{\mathrm{ref}} - m_{\theta,k}).$$

The tracker uses a bounded proportional unicycle law,

$$v_k = \mathrm{clip}(k_v \|e_k\|_2 \max(\cos\psi_k, 0), 0, v_{\max}), \qquad \omega_k = \mathrm{clip}(k_\omega \psi_k, -\omega_{\max}, \omega_{\max}).$$

The factor $\max(\cos\psi_k, 0)$ suppresses forward motion when the robot is turned away from the waypoint.

D.4 No-go barrier

The implemented no-go cost combines a logarithmic warning band with a quadratic violation term,

$$J_{\mathrm{nogo}}^{(i)} = w_{\mathrm{near}} \log\left(1 + \left(\frac{\max(b - d_{\mathrm{clr},i}, 0)}{b}\right)^2\right) + w_{\mathrm{nogo}} \left(\frac{\max(-d_{\mathrm{clr},i}, 0)}{\epsilon}\right)^2.$$

For $d_{\mathrm{clr},i} \geq b$, the cost is zero. For $0 \leq d_{\mathrm{clr},i} < b$, the log-shaped warning discourages motion close to blocked workspace geometry. For $d_{\mathrm{clr},i} < 0$, the quadratic term penalizes violations.